Homology sequence statements

These statements concern a short exact sequence of chain complexes. They do not assert a geometric Mayer–Vietoris theorem. LLM-assisted text; see PROVENANCE.md.

Result 1 (connecting then first vanishes). [Mathlib homology sequence, δ_comp]
Let K be a type. Let *field* be a field structure on K. Let S be a composable pair of chain maps with zero composite over K. Let n be a natural number. Assume S is short exact (`h`). Then

$$H_n(f(S)) \circ \delta_{n+1}(S) = 0.$$

Result 2 (last then connecting vanishes). [Mathlib homology sequence, comp_δ]
Let K be a type. Let *field* be a field structure on K. Let S be a composable pair of chain maps with zero composite over K. Let n be a natural number. Assume S is short exact (`h`). Then

$$\delta_{n+1}(S) \circ H_{n+1}(g(S)) = 0.$$

Result 3 (homology exact at first). [Mathlib homology sequence, homology_exact$_1$]
Let K be a type. Let *field* be a field structure on K. Let S be a composable pair of chain maps with zero composite over K. Let n be a natural number. Assume S is short exact (`h`). Then

$$\mathrm{exact}(\delta_{n+1}(S), H_n(f(S))).$$

Result 4 (homology exact at middle). [Mathlib homology sequence, homology_exact$_2$]
Let K be a type. Let *field* be a field structure on K. Let S be a composable pair of chain maps with zero composite over K. Let n be a natural number. Assume S is short exact (`h`). Then

$$\mathrm{exact}(H_n(f(S)), H_n(g(S))).$$

Result 5 (homology exact at last). [Mathlib homology sequence, homology_exact$_3$]
Let K be a type. Let *field* be a field structure on K. Let S be a composable pair of chain maps with zero composite over K. Let n be a natural number. Assume S is short exact (`h`). Then

$$\mathrm{exact}(H_{n+1}(g(S)), \delta_{n+1}(S)).$$